HAI Assignment 2

Submission Details

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Task Description

The task is **classification of a meme as hateful / non-hateful.**

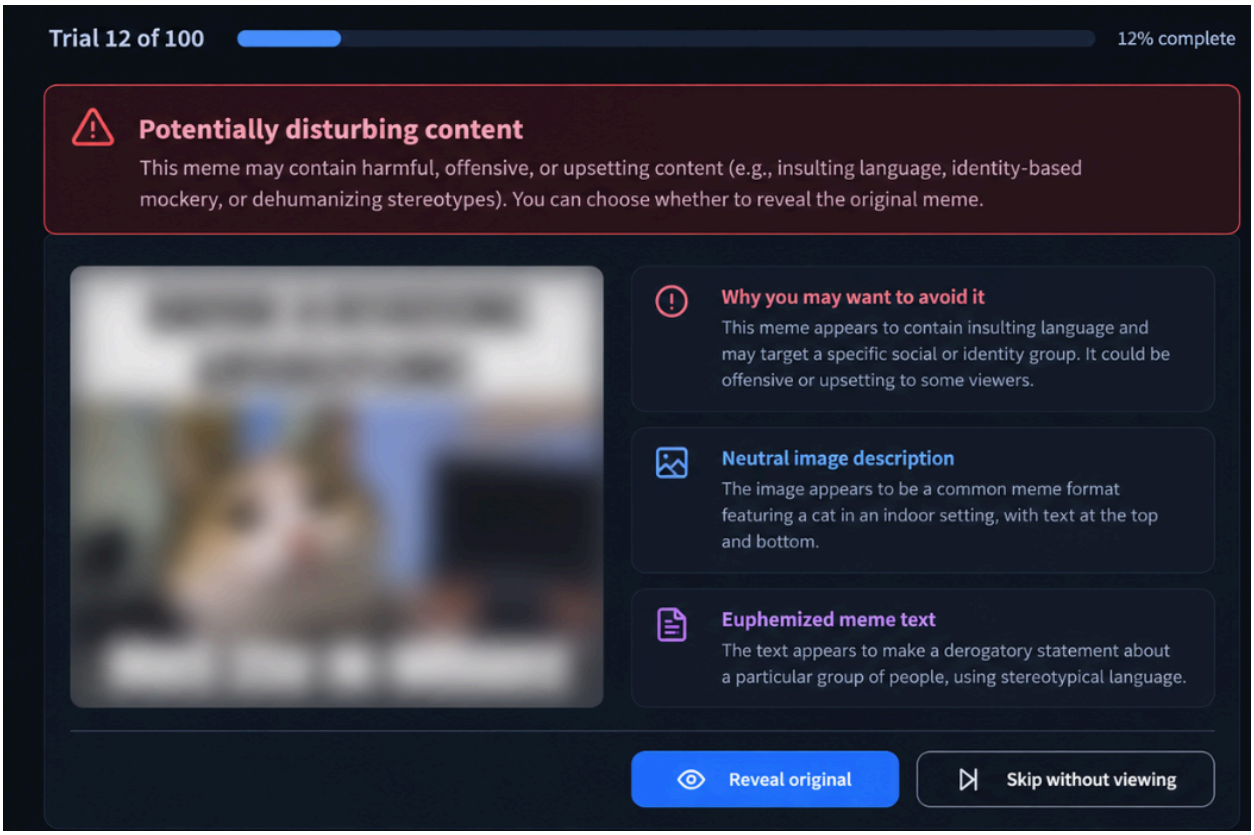
A task trial shows a meme (typically a graphic image with some text, or occasionally just text on a neutral background) to a user, and asks the user to classify the meme depending on whether the user thinks the meme has content which can be perceived as hateful to a certain individual or group of people (generally but not necessarily towards people from a particular race, ethnicity, gender, religion, political outlook). The AI will also classify the meme independently and report its confidence value

In addition, before showing a meme to a user for classification, an **AI agent will try to find specific triggers** which might upset a specific class of users. If the AI agent finds such a trigger, it will blur the image and instead provide a **textual description of the meme content** using euphemism wherever necessary and explain why it thinks viewing the meme might be upsetting to some people.

The user can choose to skip to the next suggestion or choose to unblur and rate it anyway. For every meme the user chooses to rate, they are required to:

1. Select if they think that the meme contains hateful content (on a likert scale)
2. Select if they would have preferred not having shown the meme (if it was upsetting to them)

Potentially distressing content and hate speech are distinct: a non-hateful meme can contain gore, profanity or slurs in quotation, while a hateful meme may be visually unobjectionable until its social implications are considered.



Above is a sample interface where respondents who are sensitive to stereotypical language might skip but those who are only sensitive to graphic gore or nudity may still classify it.

A skipped meme is a viewing decision. The actual human interface experiment is not part of the completed Assignment 2 model benchmark.

Dataset Description

Source: limjiayi/hateful_memes_expanded, `train.jsonl`, from Hugging Face.

The source contains meme IDs, image paths, captions and binary labels (`0` nonhateful; `1` hateful). From 12,887 eligible training records, a fixed seed (`5942026`) selected **100 examples of each label (for reproducibility)**, shuffled to make **200 unique trials**.

While the dataset represents ground truth for **hateful / non-hateful classification,** I do not have ground truth for **triggering / non-triggering** classification. I intend to get that information from conducting human trials in Assignment 3 (assuming honest reporting, anyone reporting an unblurred content as triggering can be considered as ground truth and if no one reported the content as triggering then it can be assumed to be non-triggering.

The caveat here is that people who skip a meme based on the AI's feedback cannot vote if the meme was actually triggering or not. However, the idea is that in a diverse population, for every meme I would expect at-least a few respondents to have voted. Although there might be a sampling bias depending on their outlook of what is considered as "triggering".

AI Model Description

The final model is **Gemma 4 26B A4B**, invoked via the Cloudflare Workers AI multimodal Chat Completions API with model identifier `@cf/google/gemma-4-26b-a4b-it`. It is an off the shelf vision-language model; no training or fine tuning was conducted.

The original meme image is downloaded from the verified Hugging Face record, resized for the API, and provided together with the caption. The reference label is never included in the model input. A single request returns a schema-validated JSON object with **two logically distinct sections**. The screening and classification outputs share one model invocation.

AI Model Function A

The model produces a boolean potential distress flag, content categories, specific cues, a neutral image description, and a euphemized paraphrase of the text. The

advisory describes who might prefer not to see the material by their **viewing preferences**, not by automatically attributing discomfort to an entire demographic. The prompt explicitly instructs the model to avoid judging whether the meme is hateful within this preview.

AI Model Function B

The model outputs a binary hateful/nonhateful judgment; supporting visual and textual evidence; an explanation; a possible alternative interpretation; and a self-assessed confidence score between 0 and 1. It is prompted to consider stereotypes, sarcasm, quotation and the combined meaning of image and text. **Model-generated confidence has not been calibrated** against observed correctness.

The Python implementation calls Cloudflare API, using Pillow to prepare image inputs, Pydantic to validate responses, and an append-only checkpoint to recover from API limits. A GitHub Actions workflow executed the 200 trials and committed the outputs.

Task and Model Justification

Task: Hateful meme classification

Hateful intent in memes often emerges from the interaction between images and text, making classification more challenging than conventional text classification. A meme can have completely benign text and image, however when combined together (often with some recent context that has been popular in social media) it might have an entirely different implication which can be potentially considered hate speech. Such complex interactions are often difficult to capture even using state-of-the-art AI models and needs human assistance to ensure proper classification.

However, apart from the fact that human review is not as scalable as AI, the other issue is that the respondents themselves can be sensitive to certain triggers (such as gore, past trauma, references to war crimes etc.). As the conductors of the study, it is our ethical responsibility that all respondents are not involuntarily shown such potentially triggering examples.

Since the task of detecting triggers is more straightforward that intent classification, it would be reasonable to assume that AI would do a great job as filter to decide whether to show an example to an individual respondent.

The idea is that with a diverse set of respondents, we can and should avoid showing potentially graphic or upsetting content to all users involuntarily, but some users who voluntarily choose to view it should be able to rate it so that we have a more enriched dataset. This can lead to a better approach using human-AI collaboration compared to human-only or AI-only approaches if we view the union of dataset diversity and respondent satisfaction as our optimization objective.

Model:

I chose to implement using API of an off-the-shelf model. The model should generate hateful or non-hateful classifications alongside supporting visual and textual evidence, reasoning, and confidence estimates. Structured outputs are needed to separate neutral exposure screening from hateful content classification, preventing the latter from being explicitly disclosed in sanitized previews.

The choice of model was influenced by availability of free tokens. I wanted a multimodal-to-text model with generous daily token limits to complete the study in limited time. Cloudflare's Gemma-4 was a reasonable choice in free-tier due to its generous limit of 10k neurons per day.

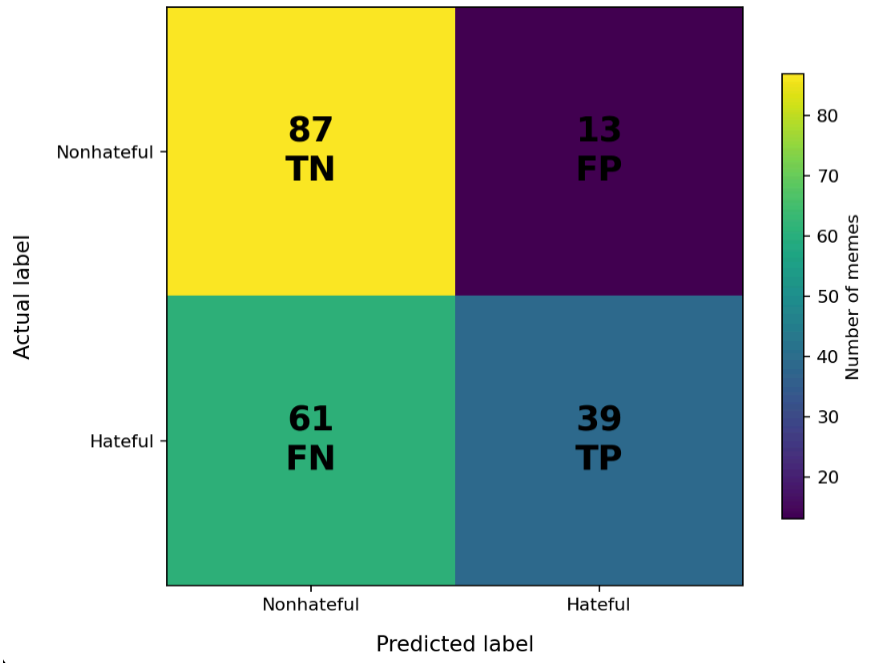
Another advantage was the ability to combine the function A and function B into one singular API call and one singular output JSON, without the need for 2 separate calls with same image.

Evaluation and Results

The binary classifier is benchmarked against the 200 original dataset labels using **accuracy, hateful precision, hateful recall, hateful F1, specificity, balanced accuracy and confusion counts**. Recall matters because undetected hateful memes are a failure of moderation; precision matters because false positives may incorrectly suppress benign or critical speech.

Measure	Observed result
API response coverage	200 / 200 (100%)

Measure	Observed result
Accuracy	**63.0% (126 / 200)**
Hateful precision	75.0%
Hateful recall	**39.0%**
Hateful F1	0.513
Nonhateful specificity	87.0%
Balanced accuracy	63.0%
Potentially distressing flag rate	44.5% (89 of 200)



Low Recall: The most concerning issue with AI-based intent classification in this case is low recall (model has a higher tendency to classify hateful memes as non-hateful, which can be potentially dangerous).

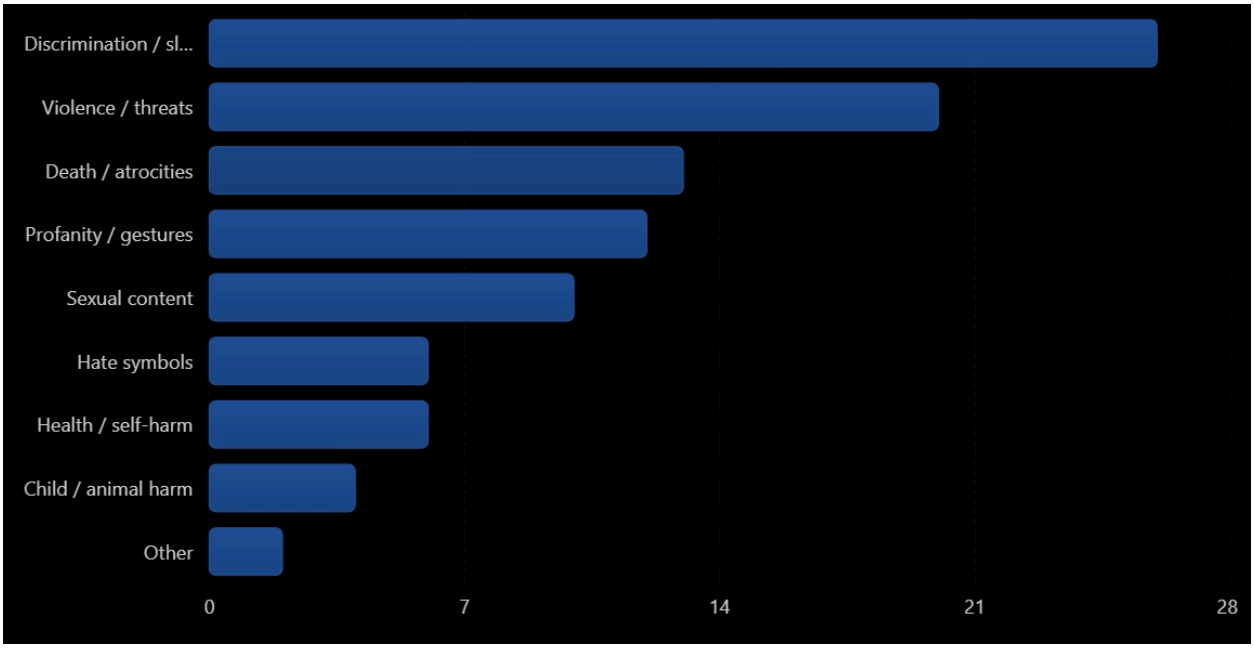
Confidence concerns: Mean self-reported confidence is approximately **0.934 on correct predictions** and **0.928 on errors**. The model assigned confidence of at least 0.8 to **73 of 74 incorrect predictions**. It is therefore particularly important that participants verify the evidence rather than trust confident labels by default.

I have also performed ROGUE analysis with text contained in the meme as ground truth and the AI description as candidate expression.

Category	Trials	ROUGE-1 F1	ROUGE-2 F1	ROUGE-L F1
All memes	200	0.674	0.517	0.664
AI flagged as potentially triggering	89	0.666	0.499	0.660
Not flagged	111	0.681	0.531	0.668
Hateful ground truth	100	0.606	0.435	0.596
Nonhateful ground truth	100	0.742	0.599	0.73

However, ROGUE is not an appropriate metric here since original captions are not human written reference paraphrases. A good euphemization may deliberately replace explicit terms, lowering ROUGE despite preserving meaning. ROUGE also cannot meaningfully assess the binary hate predictions or image descriptions without suitable textual reference outputs.

Sensitivity Trigger Alerts



Above is a classification of the trigger alerts generated (a meme can generate multiple alerts). 89 out of 200 memes (44.5%) were classified by Cloudflare

Gemma 4 as potentially triggering or distressing. The remaining 111 memes (55.5%) were not flagged. There is no established ground truth for these in the dataset yet therefore classification accuracy is hard to predict.

Human-AI Performance Predictions

As we have seen from the AI performance report, the classification accuracy is ~ 63% (which is pretty low) and false-negative rate is 39% (which is pretty high). This matches my expectation that AI alone is unable to correctly detect the subtle sarcasm, differences between word and intent, correlations between text and image, as well as references to events.

I do not currently have the numbers without human trial, but I am certain that human-only evaluation would yield a higher accuracy than AI only. But human-AI classification performance can be similar (or marginally better due to cues from AI if human is unaware of events that might provide additional context) to human-only. The benefit of using Human-AI in this process majorly comes from having the act as a filter before the trials are shown to the human classifiers, which improves user satisfaction. While I have sufficient reason to believe that this would improve user satisfaction, I would need to perform Assignment - 3 to get the exact numbers.

Acknowledgements

I acknowledge extensive usage of AI tools (OpenAI Codex) for coding assistance. Task ideation and planning as well as project structuring is done by me, and I have reviewed the AI generated code and output to ensure it matches my expectations. I also acknowledge CloudFlare's generous free access to their model Gemma 4 (26B parameters) through API.